

WAIT YOUR TURN: A QUEUE-THEORETIC FOUNDATION FOR PROCESSOR SCHEDULING

MUSTAFIF KHAN

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1. INTRODUCTION TO QUEUE THEORY

Queueing theory is the mathematical study of waiting lines or queues, the model is constructed so that queue length's and waiting time can be predicted [1]. The theory has a broad range of applications, including hospital check-up systems and call center queue management. These systems are influenced by the duration of wait times and the associated level of customer's patience. As such, optimizing and reducing wait times is essential for improving service quality. This discussion will primarily focus on another application of queue theory: processor scheduling, specifically addressing how a computer's CPU manages processes within the operating system.

1.1. Kendall's Notation.

The way that queue models are described is under Kendall's Notation which was firstly proposed in 1953 that had three factors written as $\mathbf{A/S/c}$. Firstly, $\mathbf{A}$ denotes the time between arrivals to the queue or the arrival process which for example can be Markovian, General, Degenerate distribution. Secondly, $\mathbf{S}$ denotes the service time distribution which gives the distribution of time of the service of a customer, some common ones are exponential (Markovian), deterministic (Degenerate), etc. Lastly, $\mathbf{c}$ represents the number of servers or service channels.

Kendall's Notation has since been extended to $\mathbf{A/S/c/K/N/D}$, where $\mathbf{K}$ is the capacity of the queue, $\mathbf{N}$ is the size of the population of jobs to be served, and $\mathbf{D}$ is the queue discipline. When these additional three are not specified, it is assumed, $K = \infty$, $N = \infty$, $D = \text{FIFO}$.

1.2. Single Queue Node.

A queue or queue node can be thought of as nearly a *black box*, where jobs arrive to the queue, wait some time being processed in the queue, then depart from it. Consider Figure 1 which shows an arrival stream of jobs be sent to the queue system or node, which is directed to a buffer and then a system (or can be a blackbox), then departs from the node [2].

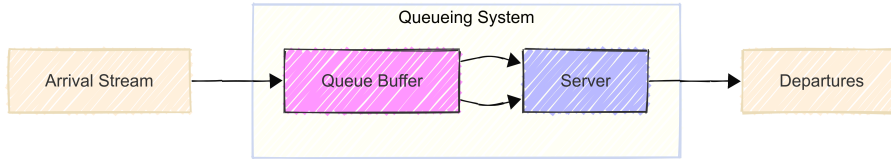


FIGURE 1. A Single Queue Node including a Buffer and Server

An analogy that's great to understand is a cashier in a grocery store, the customers are waiting in line to get their items scanned, then paid, once the customer has paid and their items packed, they depart and the next customer gets processed. In this instance each cashier processes one customer at a time, this is a queueing node with only one server. When the cashier is really busy, and the customer will

leave immediately, is referred to as a queue with no buffer (or no waiting area), while a setting with a waiting area for upto n customers is called a queue with a buffer of size n [2].

2. AN EMPIRICAL ANALYSIS OF SCHEDULER STRATEGIES
3. SINGLE VS MULTICORE PROCESSOR SCHEDULING
4. CONCLUSION

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, MOKA READS COLLECTIVE,
Email address: mustafik.khan@mokareads.org
URL: mokareads.org